Prepared By	09-Jun-2012	Mr.Chatchawan S.	RNC. Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved
1	13-Feb-2013	Mr.Chatchawan S.	Mr.Phuchin S.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-003-2012

<u>Addendum</u>

WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-006

RNC WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: **Renew Coil Engineering & Supply Co., Ltd.**

WPS No.: RNC-WPS-006 | Date: 13-Feb-13 | Revision No.: 1

Supporting PQR No.(s): RNC-PQR-006 | Date: 13-Feb-13 | Type(s): Manual

Welding Process(es): Gas Tungsten Arc Welding + Shield Metal Arc welding (GTAW +SMAW)

JOINT (QW-402)

Joint Design: **Single V**

Backing (Yes) ____ (No) X

Backing Material (Type): N/A

☐ Metal ☐ Nonfusing Metal ☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. 1 Group No. 1,2 to P-No. 1 Group No. 1,2

Material Specification Type and Grade: SA 333 Gr.1,6 SA420 WPL6 TO SA 333 Gr.1,6 SA420WPL6

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal : Groove 1.6 mm - 14.22 mm. Fillet All

Pipe Dia. Range : Groove All Fillet All

Other

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.28	5.5
AWS No. (Class)	ER-70S-G	E8016-G
F-No.	6	4
A-No.	1	1
Size of Filler Metals	Ø 2.0 & 2.4 mm	Ø 3.2 mm.
Deposited Weld Metals	2.0 mm	5.11 mm.
Thickness Range :		
Groove	4 mm Max.	10.22 mm. Max.
Fillet	All	All
Electrode-Flux (Class)	-	
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco TGS- 1N or Equivalents	Kobelco NB- 1 or Equivalents

POSITION (QW-405)

Position(s) of Groove: ALL

Welding Progression: Up X Down

Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: 593 - 649°C

Time Range: Holding Time 1 Hour

Notes: Heating & Cooling Rate = 200 ° Max.

PREHEAT (QW-406)

Preheat Temp. Min. 10°C

Interpass Temp. Max. 238°C

Preheat Maintenance: N/A

Notes

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

RNC

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-006 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current [X] AC [X] DC Polarity EN(GTAW)+ EP(SMAW)

Amps (Range) 90-140(GTAW) + 80-130(SMAW) Volts (Range) 9-13(GTAW) + 18-24(SMAW)

Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS

Orifice or Gas Cup Size 8.0 mm - 19.0 mm

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE / MULTIPLE ELECTRODE

Travel Speed (Range) 6.0-10.0 Cm/Minute

Peening N/A

Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	90-130	9-12	6.0-10.0	
2nd	GTAW	ER 70S-G	2.4	DC EN	100-140	10-13	6.0-10.0	
3 & Over	SMAW	E8016-G	3.2	AC/DCEP	80-130	18-24	6.0-10.0	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2007 latest Revision.**

QA/QC RNC

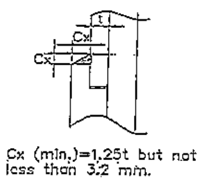
Prepared by : Mr. Chatchawan Name Signed Date 13/03/13.

Reviewed by : ภูชิน สีหอมไชย Phuchin Seehomchai Name Signed Date 22/03/13

Approved by: ภูชิน สีหอมไชย Phuchin Seehomchai Name Signed Date 22/03/13.

Sheet 2 of 2

ATTACHED FIGURE JOINT SKETCH



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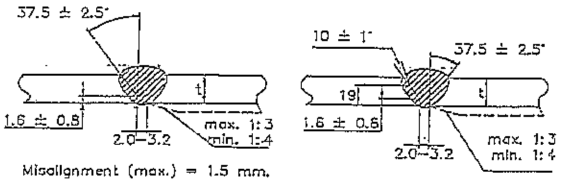

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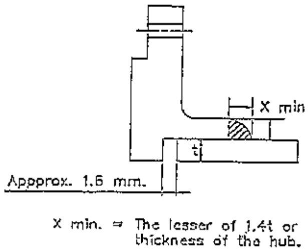

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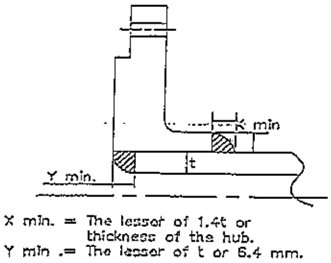

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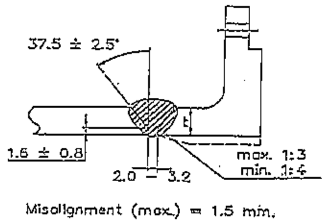

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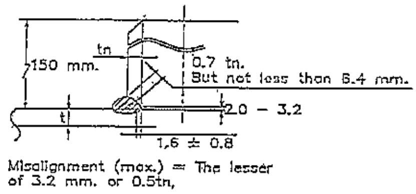

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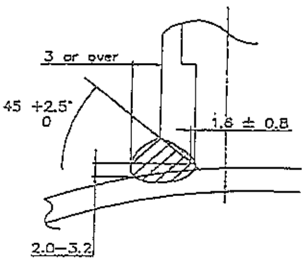

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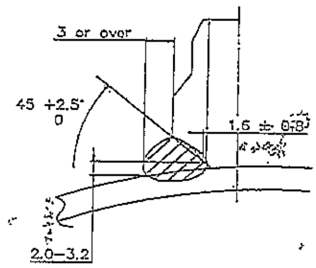

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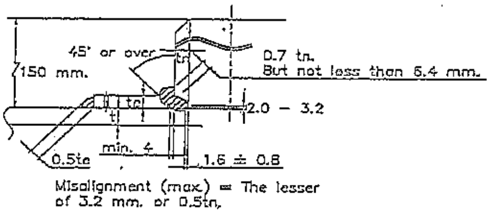

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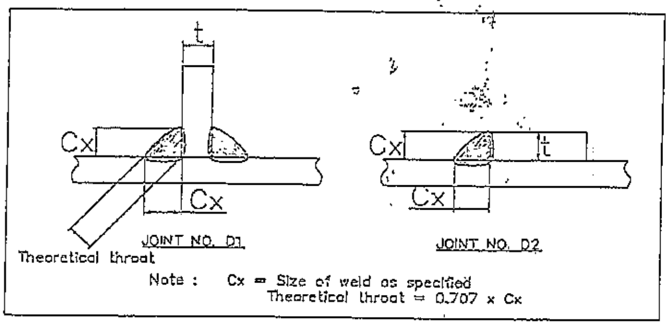


Procedure Qualification Record (PQR)

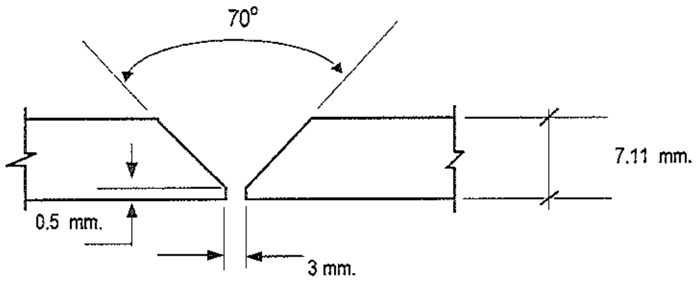
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-006

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-006	Date :	13 February 2013
WPS Number :	RNC-WPS-006		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic):	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)

Material Spec.	A333	To	A333
Type or Grade	Gr.6	To	Gr.6
P. No.	1	To	1
Thickness of Test Coupon	7.11 mm.		
Diameter of Test Coupon	ø 6"		
Others			

FILLER METALS (QW-404)

	GTAW	SMAW
SFA Specification	A 5.28	A 5.5
AWS Classification	ER70S-G	E8016-G
Filler Metal F-No.	6	4
Weld Metal Analysis A-No.	1	1
Size of Filler Metal	Ø 2.4 mm.	Ø 3.2 mm.
Others	Kobelco TGS-1N	Kobelco NB-1
Weld Metal Thickness	-	

POSITION (QW-405)

Position of Groove	6G
Progression of Welding (Uphill or Down)	Up Hill
Others	N/A

PREHEAT (QW-406)

Preheat Temperature	35° C
Interpass Temperature (max.)	180° C
Others	N/A

POSTWELD HEAT TREATMENT (QW-407)

Temperature	610° C
Hold Time	1 Hour
Others	Cooling & Heating Rate 120° C - 190° C (MAX)

GAS (QW-408)

Percent Composition

	Gas (es)	Mixture	Flow Rate
Shielding	Argon	99.95%	15 L/Min.
Trailing	-	-	-
Backing	-	-	-

ELECTRICAL CHARACTERISTICS (QW-409)

Current	DC (GTAW)		DC (SMAW)
Polarity	EN		EP
Amperage	110 - 130 (GTAW)	Voltage	10 - 13 (GTAW)
	84 - 86 (SMAW)		24 (SMAW)
Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
Others	N/A		

TECHNIQUE (QW-410)

Travel Speed	4.61 - 8.91 cm/min.
String or Weave Bead	Both
Oscillation	N/A
Multiple or single Pass	Multiple Pass
Multiple or Single Electrodes	Multiple Electrodes
Others	N/A

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542
69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: RNC-PQR-006

Tensile Test Report No.TS-13-02-085

Specimen	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (KN)	Ultimate Strength (N/mm^2)	Type of Failure & Loc.
546/13 TR1	19.11	6.93	132.43	58.87	444.56	OUT OF WELD
546/13 TR2	19.13	6.76	129.32	59.06	456.67	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-034

Type and Figure Number	Type of Bend	Indication	Results
546/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
546/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
546/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
546/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

Other Tests

Type of Test
- Radiographic examination results Accepted Report No.WQT-008,011
- Hardness Testing Report No. HV-13-02-016
- Macrostructure Report No. MA-13-02-050
- Impact Testing Report No.IM-13-03-005 (-60°C)

Welders Name Mr.Panja S. Welder Number W-006

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date : 22/03/13

RNC RENEW CONS ENGINEERING & SUPPLY CO.,LTD. QC

Inspected by : Mr.Wichit K.

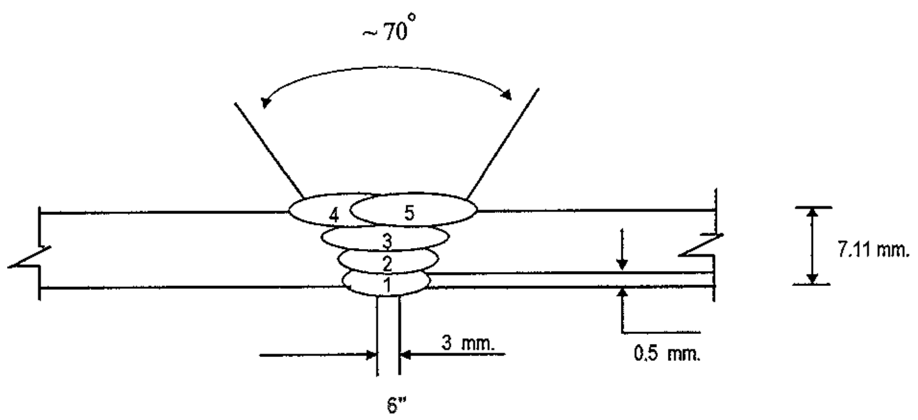
Certified by : Mr.Chaloemkiet R.

Date : 13 March 2013

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Welding Sequences for RNC-PQR-006 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	110	11	DCEN	-	-	4.61	15.75	15	-
2	GTAW	ER70S-G	2.4	130	13	DCEN	-	-	6.76	15.00	15	-
3	SMAW	E8016-G	3.2	86	24	DCEP	-	-	8.70	14.24	-	-
4	SMAW	E8016-G	3.2	84	24	DCEP	-	-	8.76	13.18	-	-
5	SMAW	E8016-G	3.2	86	24	DCEP	-	-	8.91	13.90	-	-

RNC ENGINEERING & SUPPLY CO.,LTD. QA/QC KNC

PAE Technical Service Company Limited
69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Public Registration No. (5) 1571/2542

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

REPORT No.	WQT-011	RADIOGRAPHIC EXAMINATION REPORT	INSPECTION & TEST RECORDS
DWG.No.	-		
DATE :	15-Feb-13		

CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	Welder Supply

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME		4 min. 15 sec.		DISTANCE	6" Inch. / - mm.
RADIATION SOURCE	X-ray (-) MAX - kVG. γ - ray (Ir-192)	DENSITY RANGE		2.0 - 3.5		TYPE OF PENETRAMETER	ASTM-1B
		SENSITIVITY		2%			
SOURCE RANGE	7.29 Ci.	FILM	TYPE	KODAK " MX " (Type I)		FILM PROCESSING	AUTO -
SOURCE SIZE	3 x 3 mm.		SIZE	3.5" x 8.5"			MANUAL ✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN	FRONT	0.005 Inch.		MATERIAL	A333 Gr.6
TUBE CURRENT	- mA		BACK	0.005 Inch.		THICKNESS	7.11 mm.
APPLICABLE STANDARD :	ASME Section V Article 2					ACCEPTANCE CRITERIA :	ASME-IX

RADIOGRAPHIC TECHNIQUE : ☐ SWSI ☑ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Thk(mm)	Welder No.	Film No.	Interpretation	Results	Remarks
PQR / GTAW+SMAW / WPS No.RNC-006								
MR.PANJA S.	6G	6"	7.11	W-006	A-B	NAD	ACC	After PWHT
					B-C	NAD	ACC	
					C-D	NAD	ACC	
					D-A	P ø 1 mm.	ACC	

Total 3.5"x8.5" 4 Films

3.5"x17" - Films

ABBREVIATIONS :

ACC : Accepted	C : Crack	LF : Lack Of Fusion	CRP : Concave Root Penetration
REJ : Rejected	P : Porosity	IP : Incomplete Penetration	ERP : Excess Root Penetration
SF : Surface Flaw	UC : Undercutting	TI : Tungsten Inclusion	NAD : No Apparent Defects
	SL : Slag Inclusion	BT : Burn Through	RS : Reshoot

PAE INSPECTED BY	REVIEWED BY	APPROVED BY
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Sayam S.	NAME : mr.Chatchaman S	NAME :
DATE : 16-Feb-13	DATE : 19/02/13	DATE :

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

LPN PLATE MILL

29781

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1

Report No. TS-13-02-085

Revision No. 0

Test Product	WPS No. RNC-WPS-006
Project Name	WELDER SUPPLY AND PQR
Project No.	N/A
Material Specification	A333 Gr.6
Dimension	Ø 6" (7.11 mm Thk.)
Date Received	19 February 2013
Date Tested	20 February 2013

Material Characterization	Pipe
Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Standard of Test	ASME Sec. IX : 2010 edition
Type of Test	Reduced Section Tension
Equipment's Capacity	100 TONS
Equipment / Serial No.	INSTRON / 1000DXR1414
Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	546/13 TR1	N/A	6.93	19.11	132.43	N/A	N/A	N/A	58.87	444.56	N/A	OUT OF WELD
2	546/13 TR2	N/A	6.76	19.13	129.32	N/A	N/A	N/A	59.06	456.67	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Panja S.

ID No. 3 2105 00456 92 8

Welder No. : W-006

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

LPN PLATE MILL

29787

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-034

Revision No. 0

Test Product	WPS No. RNC-WPS-006	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" (7.11 mm Thk.)	Diameter of Meudrel	28 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	546/13 BF1	Transverse face bend	7.05x37.90x160.00 mm	No Open Discontinuity	N/A	-
2	546/13 BF2	Transverse face bend	7.00x37.73x160.30 mm	No Open Discontinuity	N/A	-
3	546/13 BR1	Transverse root bend	7.01x37.70x160.16 mm	No Open Discontinuity	N/A	-
4	546/13 BR2	Transverse root bend	7.00x37.66x161.73 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Panja S.

ID No. 3 2105 00456 92 8

Welder No. : W-006

Reported by : N. [signature]

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : N. [signature]

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

13233

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

LPN PLATE MILL

Hardness Testing Report

Report No. HV-13-02-016

Revision No. 0

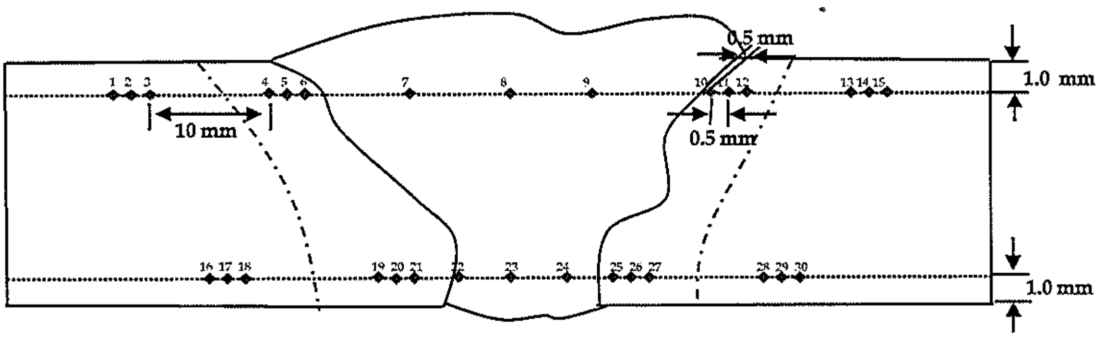
Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. RNC-WPS-006	Material Characterization	Pipe
Test No.	546/13 HN	Test Method	ASTM E92-82 (Reapproved 2003)$^{\epsilon 2}$
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Secction IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333 Gr.6	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	6" (7.11 mm Thk.)	Indentor Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Jiont	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	127.4	16	BASE	128.0	**************************		
2	BASE	128.1	17	BASE	128.9			
3	BASE	128.1	18	BASE	130.1			
4	HAZ	152.6	19	HAZ	143.1			
5	HAZ	197.6	20	HAZ	150.0			
6	HAZ	200.1	21	HAZ	154.4			
7	WELD	218.3	22	WELD	148.8			
8	WELD	208.7	23	WELD	149.4			
9	WELD	200.9	24	WELD	147.2			
10	HAZ	179.2	25	HAZ	148.6			
11	HAZ	156.9	26	HAZ	137.6			
12	HAZ	144.1	27	HAZ	136.4			
13	BASE	129.1	28	BASE	125.8			
14	BASE	128.3	29	BASE	125.6			
15	BASE	128.6	30	BASE	125.0			

Additional details : -

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 22/02/13

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 22/02/13

LPN PLATE MILL

Notes

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LPN

Testing Laboratory LPN Plate Mill Public Company Limited
199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.
Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th 13697

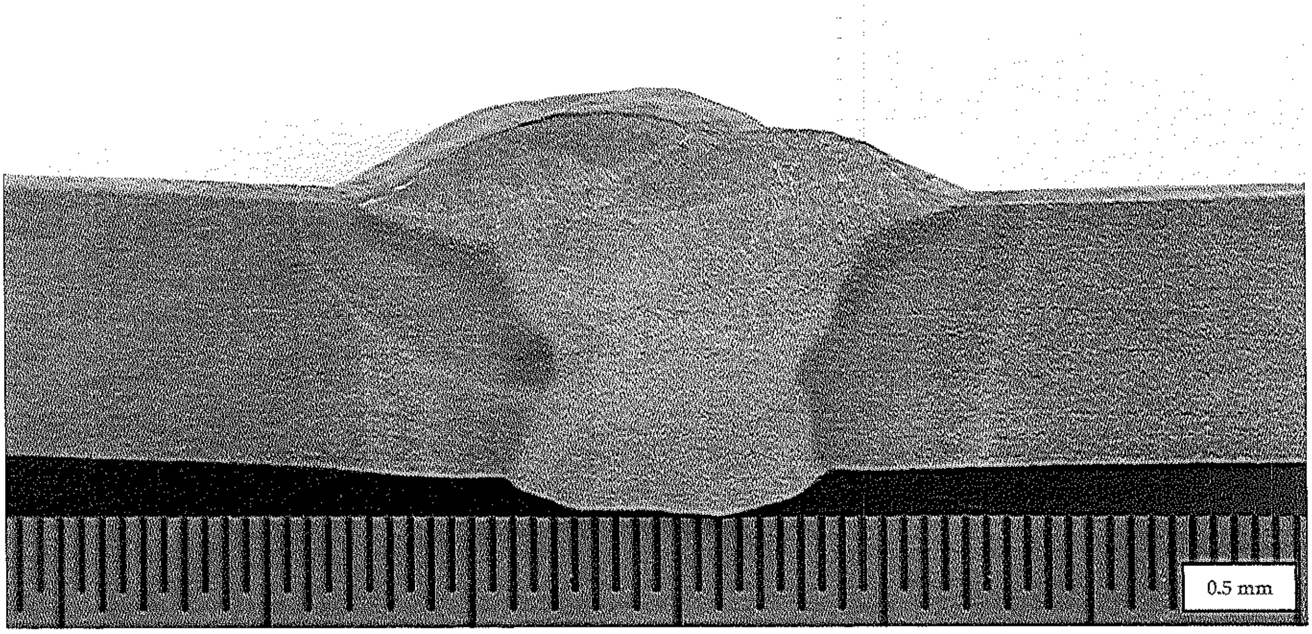
Macrostructure Analysis Report

Report No. MA-13-02-050
Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. RNC-WPS-006
Test No.	546/13 MA
Test Location	-
Material Specification	A333 Gr.6
Dimension	6" (7.11 mm Thk.)
Welding Process/Position	GTAW+SMAW/6G
Type of Joint	Butt Jiont
Welder Name / No.	Mr. Panja S. / W-006
Date Received	18 February 2013

Material Characterization	Pipe
Equipment used	Microscope upto 40X (times)
Serial No.	LEICA 0650854912
Method of Preparation	ASME Secction IX : 2010 edition
Test Method	ASME Secction IX : 2010 edition
Standard of Test	ASME Secction IX : 2010 edition
Etchant Reagent	25 ml HNO_3 + 75 ml H_2O
Time of Etching	5 minutes
Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 22/02/13

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

28608

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-03-005

Revision No. 0

Test Product	WPS No. RNC-WPS-006	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a$^{\varepsilon 1}$
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" (7.11 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	28 February 2013	Equipment 's Capacity	542 Joules
Tested Date	04 March 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	546/13-1	-60°C	5.016	10.010	54.85	104.81	117.78	Base Metal	-
2	546/13-2	-60°C	5.014	10.012	54.85	113.33			
3	546/13-3	-60°C	5.011	10.012	54.86	135.21			
4	546/13-1	-60°C	5.015	10.014	54.91	105.52	115.10	HAZ	-
5	546/13-2	-60°C	5.013	10.013	54.98	123.36			
6	546/13-3	-60°C	5.011	10.014	54.97	116.42			
7	546/13-1	-60°C	5.016	10.014	54.99	53.56	74.43	Weld Metal	-
8	546/13-2	-60°C	5.014	10.014	54.99	96.86			
9	546/13-3	-60°C	5.014	10.011	54.91	72.88			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Panja S.

ID No. 3 2105 00456 92 8

Welder No. : W-006

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date ...13 / 03 / 13...

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date ...13 / 3 / 13...

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC

LPN PLATE MILL

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

K Y Tech CO., LTD.

1/132 Moo 1 T. Phra, A. Banchang, Rayong 21130

POST WELD HEAT TREATMENT REPORT

CLIENT	: RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.	**CONTRACTOR**	: KY Tech Co.,Ltd.
PROJECT	: WELDER SUPPLY AND PQR	**DATE OF PWHT**	: 14 February 2013
TYPE OF WORK	: PWHT (LOCAL)	**REPORT NO.**	: PWHT-002
MATERIAL	: A333 Gr.6	**RECORDER NO.**	: T9YGK016
LOCATION	: KY Tech Shop	**CHART SPEED**	: 25 mm/Hr

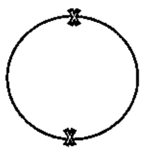
RESULTS RECORDER CHART TO BE ATTACHED

DESCRIPTION	REQUIRE		ACTUAL	
UNRESTRICTED TEMP TO	300	(°C)	300	(°C)
HEATING RATE (MAX)	200	(°C/ HR)	190	(°C/ HR)
HOLDING TEMP (MIN-MAX)	593-649	(°C)	610	(°C)
HOLDING TIME (MIN)	1	HR	1	HR
COOLING RATE (MAX)	200	(°C/ HR)	120	(°C/ HR)
COOL UNDER INSULATION TEMP FROM	300	(°C)	300	(°C)

Line No./DWG No.	Joint	Size	Thickness.	T/C	REMARK
	No.	mm	mm.	No.	
PQR		6"	7.11	1,2	
Total	1	Joint	6	DB	

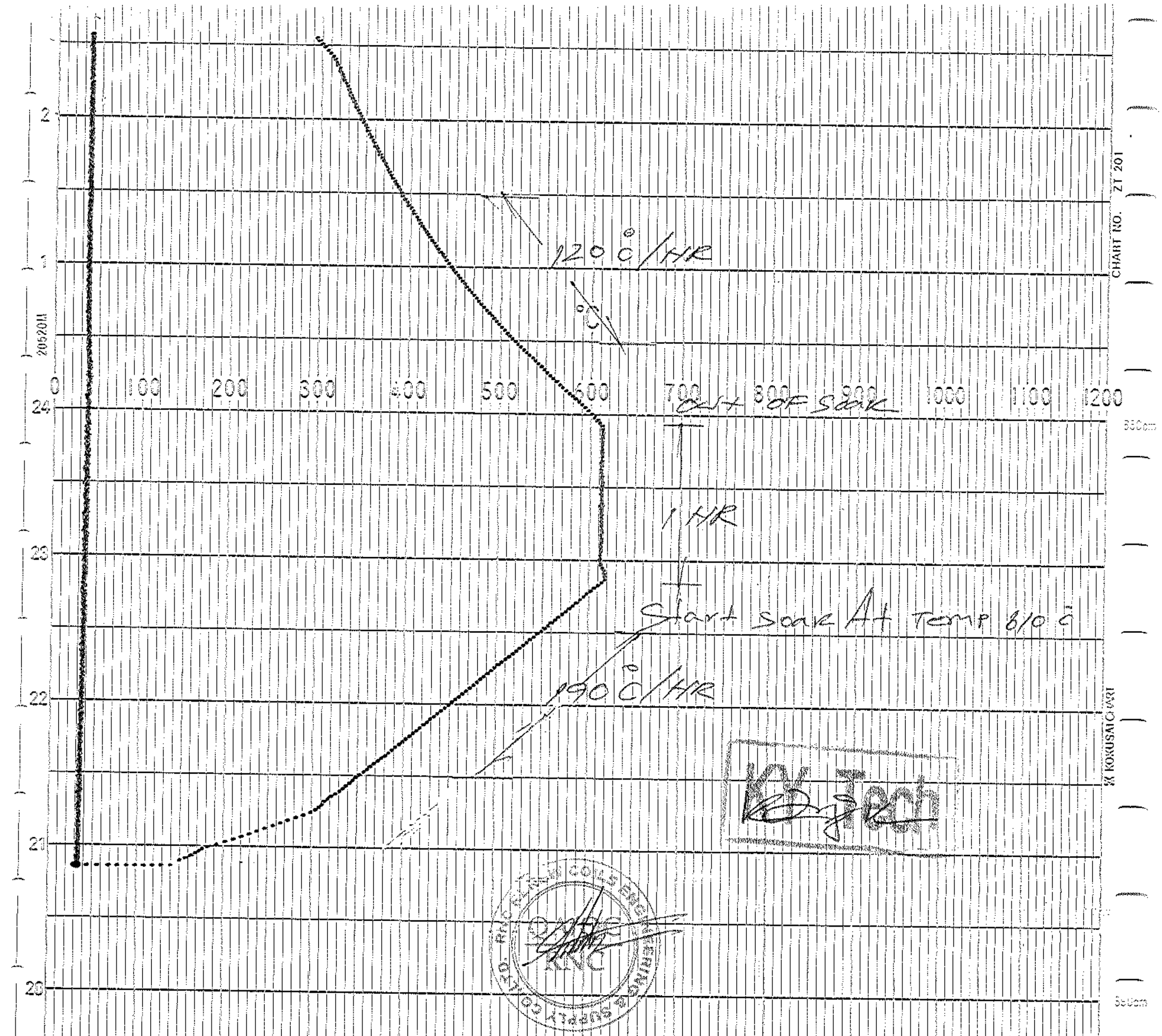
Thermocouple Location

TC#1



TC#2

	CHECKED BY	REVIEWED BY	APPROVED BY
COMPANY	K Y Tech Co., Ltd	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.	
SIGN			
NAME	MR.PAYOM KONGPOO	Mr. Chatchawan	ภูชิน สีหอมไชย
DATE	15 February 2013	11/03/13	Phuchin Seehomchai



DATE : 14 February 2013			REPORT NO. : PWHT-002			
RECORDER NO. : T9YGK016			CHART SPEED : 25 MM/HR			
DWG No. & LINE No.	Joint No.	Size (mm)	Thickness (mm)	T/C No.	COLOUR	
PQR	°C	6"	[illegible]	1,2	Red,Black	

0 100 200 300 400 500 600 700 800 900 1000 1100 1200